For your Fake Currency Detection using Deep Learning project, you can use the following Python libraries to load currency images, preprocess data, train a deep learning model, and detect potentially fake currency notes.

1. Python libraries required

Library	Purpose
TensorFlow	Build and train the deep learning model.
Keras	Create a CNN (Convolutional Neural Network).
OpenCV (cv2)	Read and process currency note images.
NumPy	Perform numerical calculations on image data.
Pandas	Load and manage dataset information.
Matplotlib	Display images and training graphs.
Seaborn	Visualize model performance and confusion matrices.
Scikit-learn	Split datasets and evaluate model accuracy.
Pillow (PIL)	Open, resize, and convert images.

2. Install all libraries

Run this command in your Jupyter Notebook, Google Colab, or terminal:

!pip install tensorflow opencv-python numpy pandas matplotlib seaborn scikit-learn pillow

3. Import the libraries in Python

import tensorflow as tfimport cv2import numpy as npimport pandas as pdimport matplotlib.pyplot as pltimport seaborn as snsfrom PIL import Imagefrom sklearn.model_selection import train_test_splitfrom sklearn.metrics import (accuracy_score, classification_report, confusion_matrix)print("All libraries imported successfully!")print("TensorFlow version:", tf.__version__)

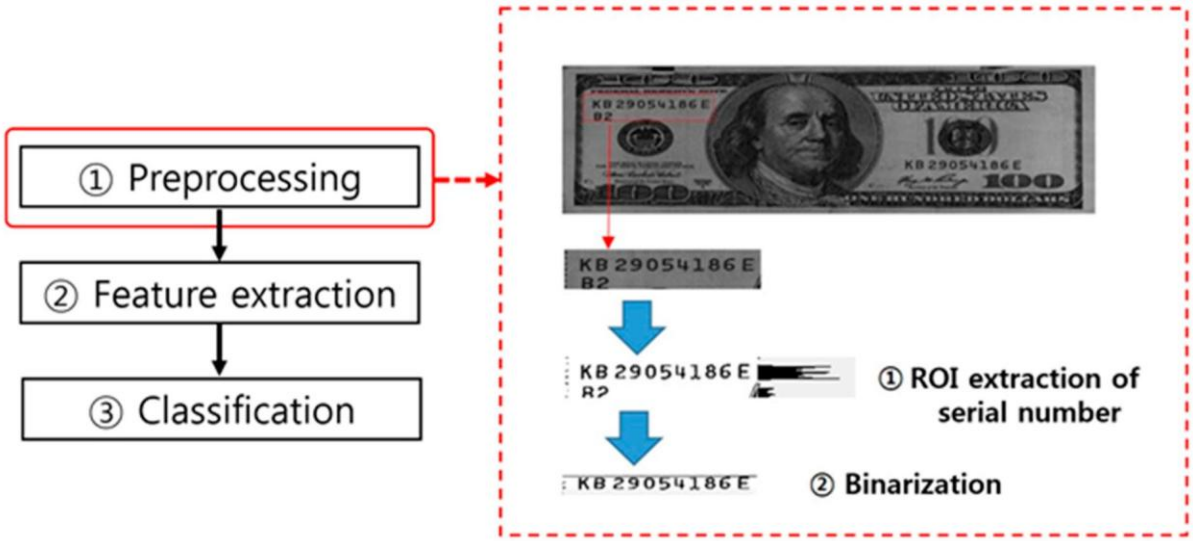
4. How the libraries are used in your project



Step 1: Dataset collection

Pandas, OpenCV, Pillow

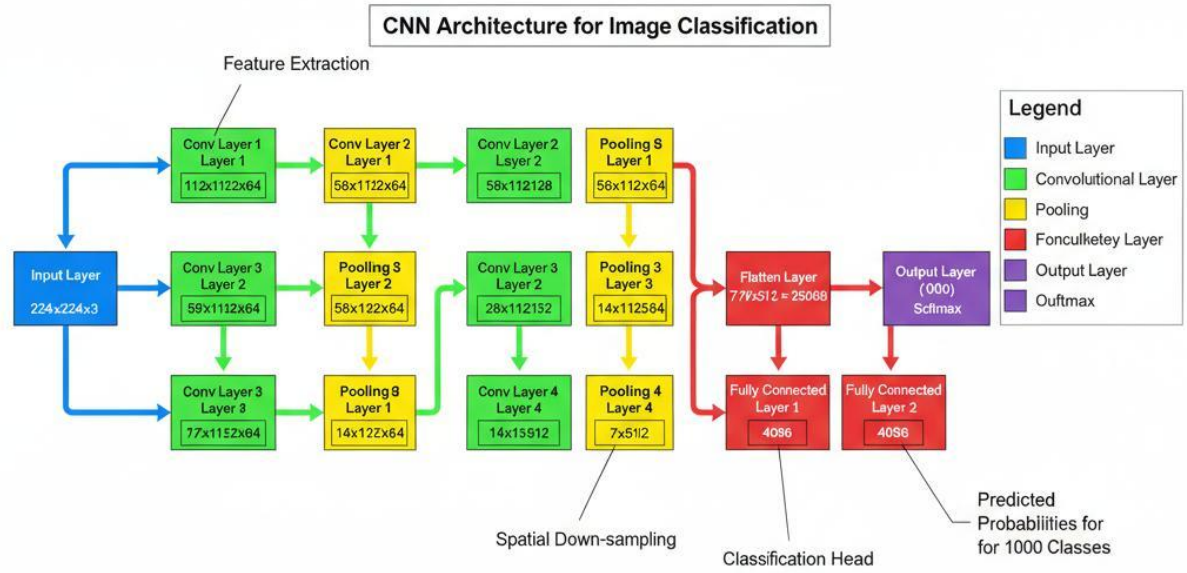
Load genuine and counterfeit-labelled banknote images.



Step 2: Data preprocessing

OpenCV, NumPy

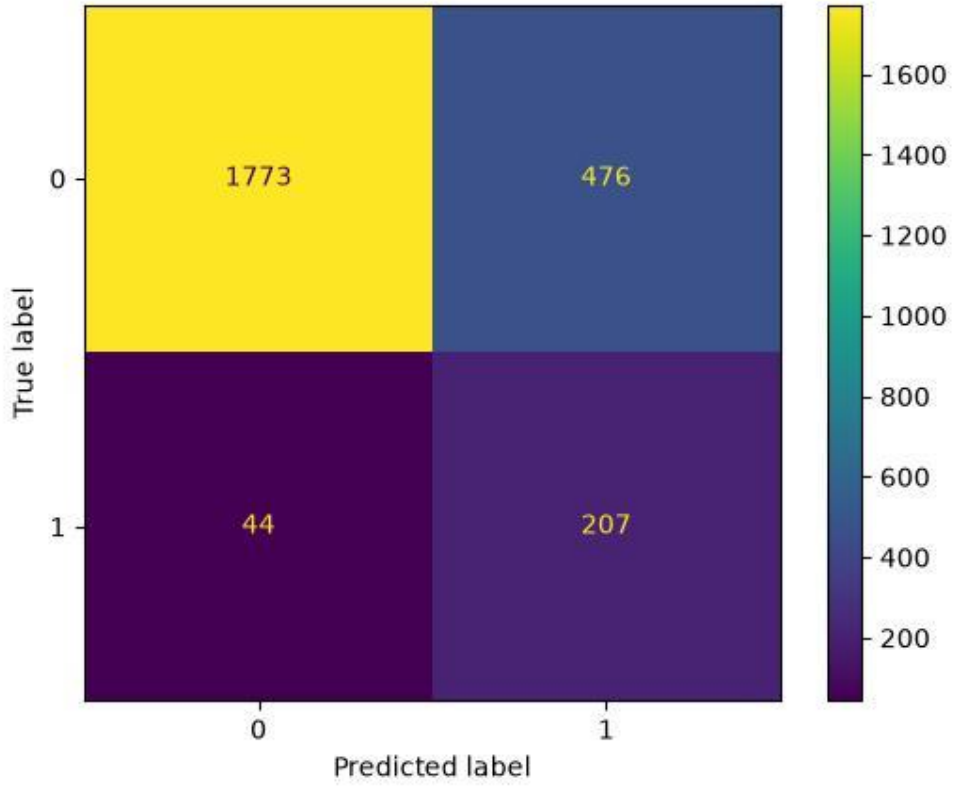
Resize images, normalize pixel values, and prepare the data.



Step 3: Model training

TensorFlow, Keras

Train a CNN to classify images using labelled examples.



Step 4: Model evaluation

Scikit-learn, Matplotlib, Seaborn

Evaluate accuracy, precision, recall, and the confusion matrix.

5. Recommended library combination

For a college-level project, start with these five:

- TensorFlow/Keras — model training
- OpenCV — image processing
- NumPy — numerical operations
- Matplotlib — visualization
- Scikit-learn — evaluation

Important: Your model's reliability depends on the quality and diversity of your labelled dataset. Image classification alone cannot conclusively authenticate a banknote; real-world verification may require security-feature inspection and expert validation.